



北京理工大学
BEIJING INSTITUTE OF TECHNOLOGY

Reversed-Chain Whole-Body Control for Robotic Manipulation on Flexible Supports

Reported by Wenka Song
Date July 15, 2026



I

Introduction

II

Method

III

Results

IV

Conclusion and ongoing work



I

Introduction

II

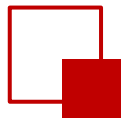
Method

III

Results

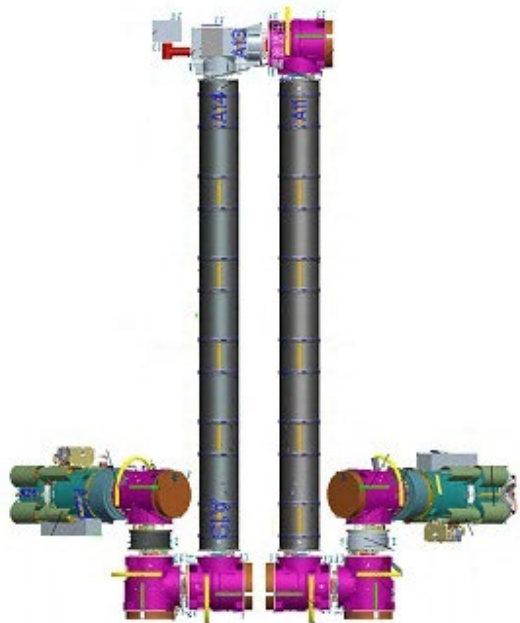
IV

Conclusion and ongoing work

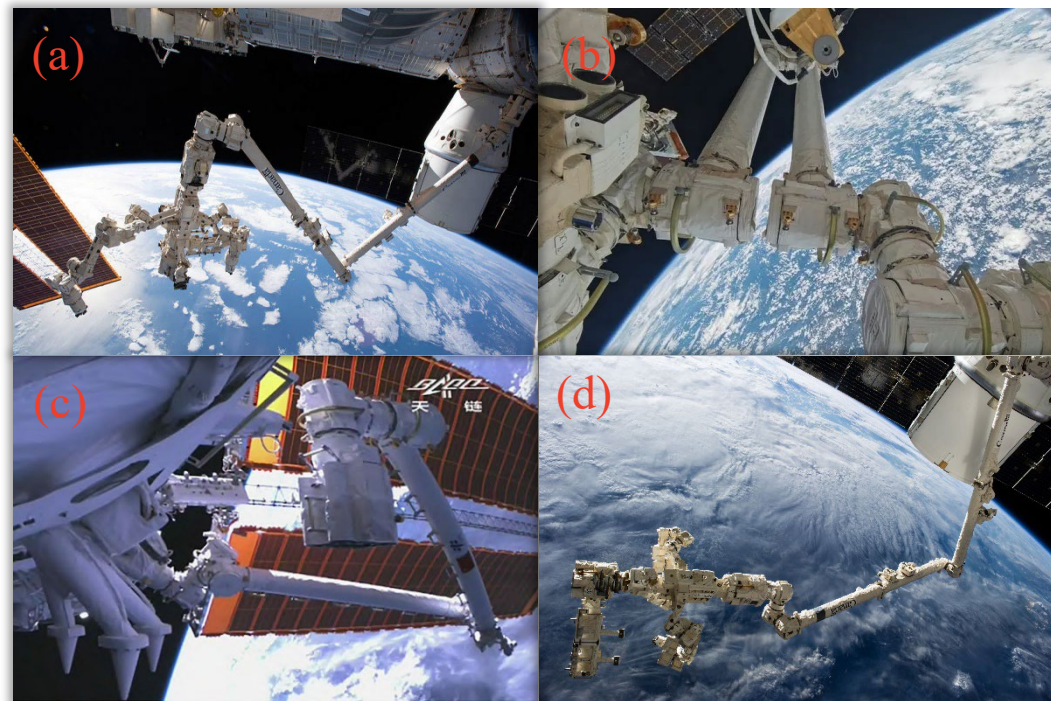


Introduction

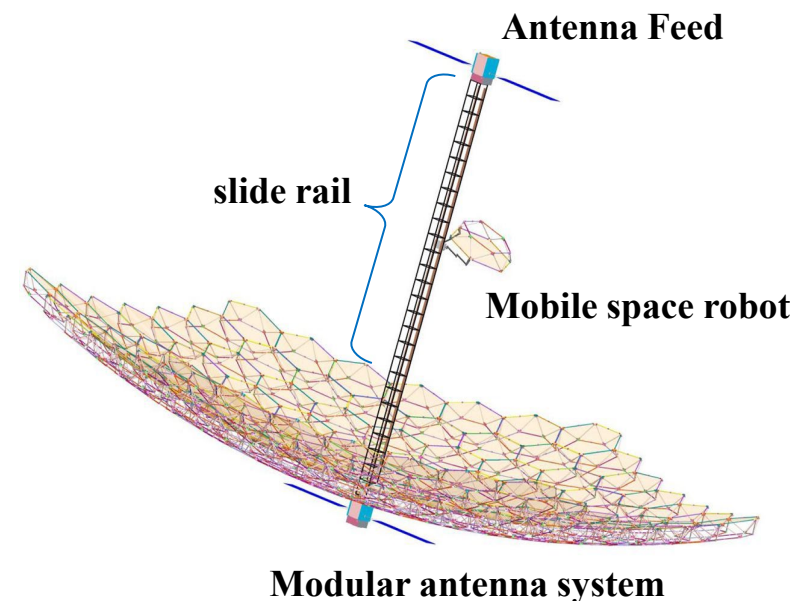
CSS robotic arm



Real scenarios of space robots



On-orbit assembly

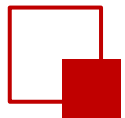


Background:

- ❑ OSAM for Large-Scale Space Structures
- ❑ Dual-Arm Synergistic Advantage
- ❑ Redundant DOF empowers optimization

Challenge:

- ❑ Strong rail–robot dynamic coupling
- ❑ Cascaded amplification of base oscillations
- ❑ Limited stability of compliant operation



Introduction

Related literature I

Given by $\hat{e}(t+i) = \hat{p}_B^D(t+i) - p_B^E(t)$

$$\hat{e}(t+i) - \Delta t \sum_{j=0}^{i-1} v(t+j) = 0$$

Optimization: $\min_u \left(\|\hat{e} - Au\|_G^2 + \|u_0 - Bu\|_Q^2 \right)$

Integrator:
$$A = \Delta t \begin{bmatrix} I & & \\ \vdots & \ddots & \\ I & \dots & I \end{bmatrix} \in \mathbb{R}^{6N \times 6N}$$

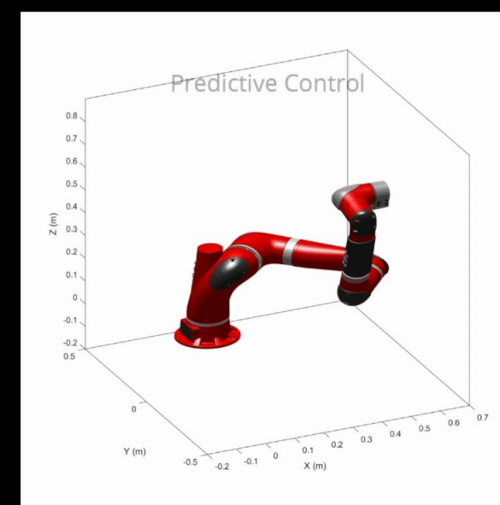
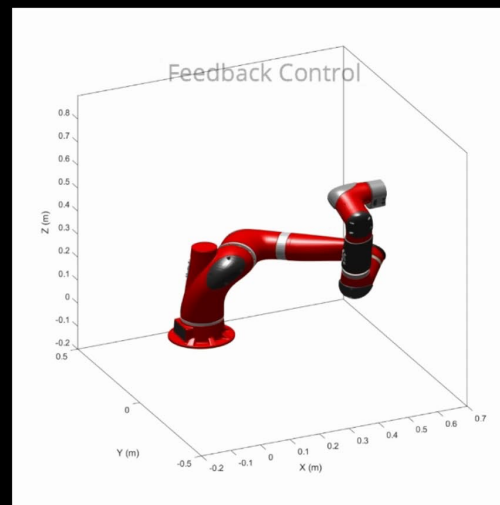
Differencer:
$$B = \begin{bmatrix} I & & & \\ -I & I & & \\ & \ddots & \ddots & \\ & & -I & I \end{bmatrix} \in \mathbb{R}^{6N \times 6N}$$

其中:

AR Model:
$$\hat{y}(t) = \sum_{i=1}^p \alpha_i y(t-i)$$



Purpose: Predict the future pose of the base

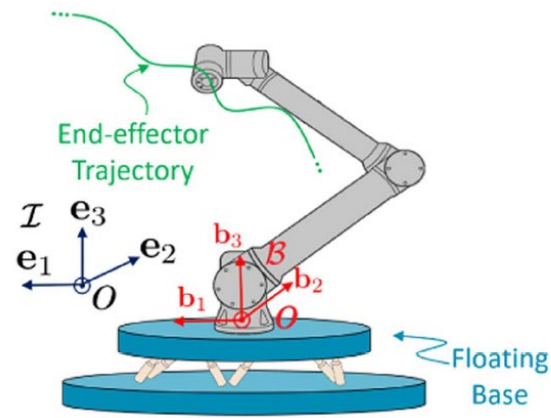


Conventional control methods, such as **feedback control** (left), make it difficult for mobile manipulators to execute accurate tasks when under **disturbance**.



Introduction

Related literature II



$$\dot{x}_e = J_b \begin{bmatrix} v_0 \\ \omega_0 \end{bmatrix} + J_m \dot{\Theta}$$

disturbance (uncertainty)

ELOVER

Nonlinear Disturbance Observer

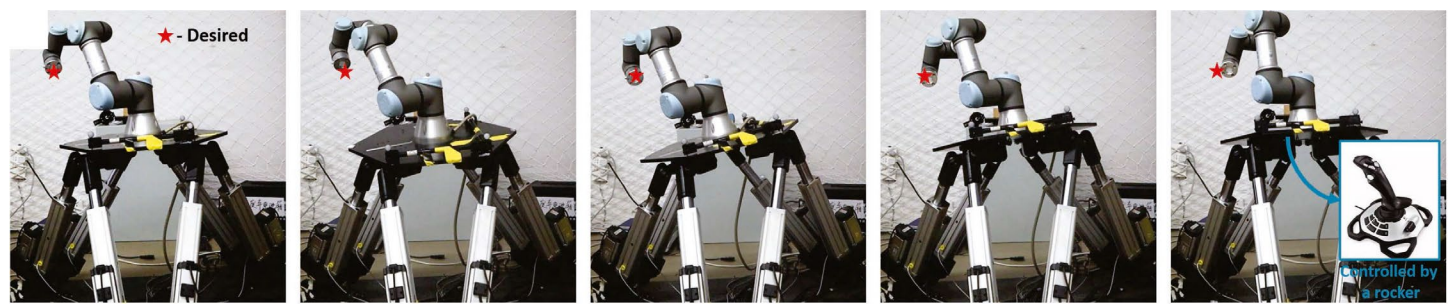
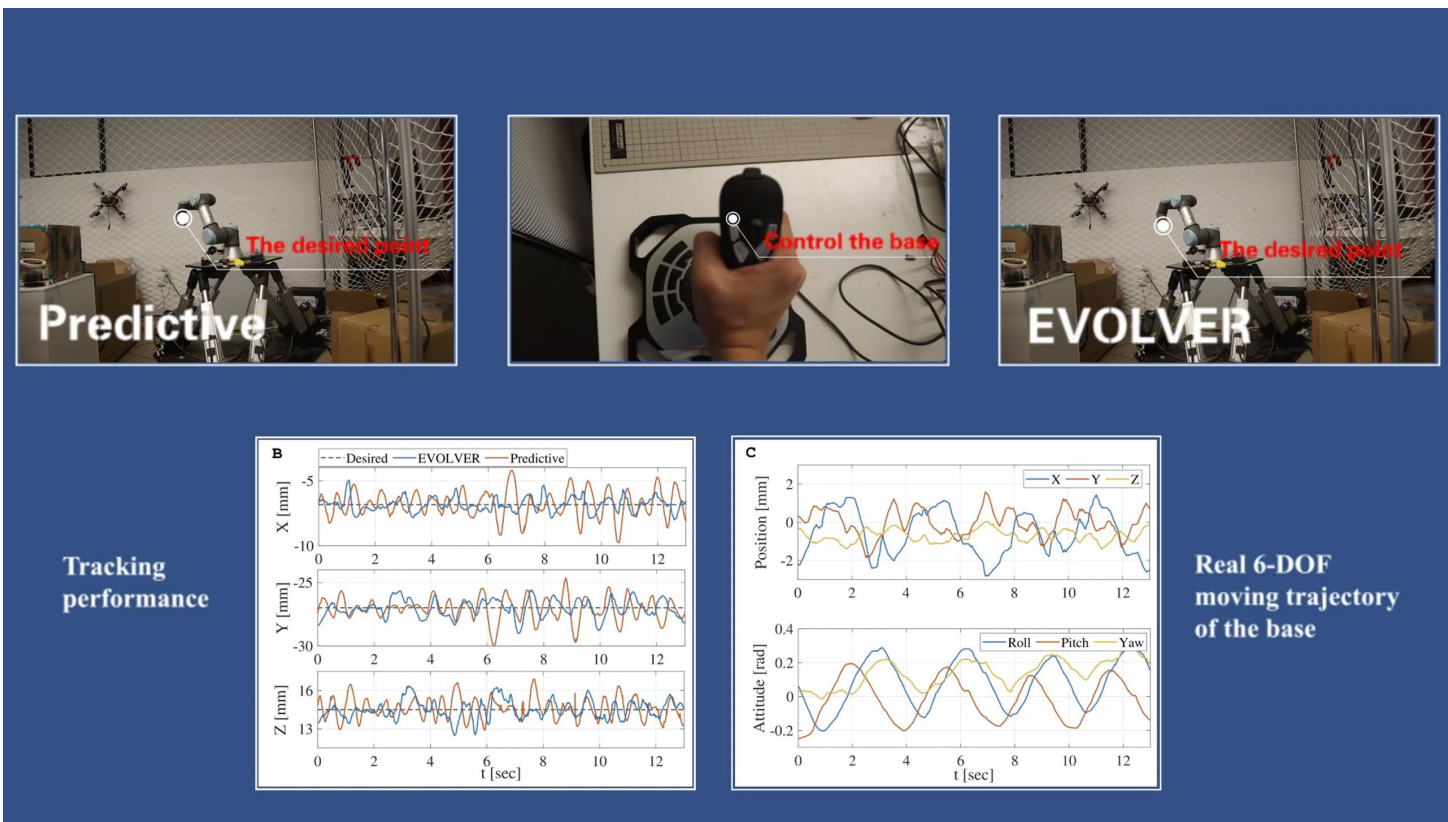
Koopman Operator

Advantages: Fast response and strong robustness

Limitations: Limited accuracy and inherent lag

Continual Learning, Adaptation, and Evolution

High Accuracy and Delay-Free Prediction

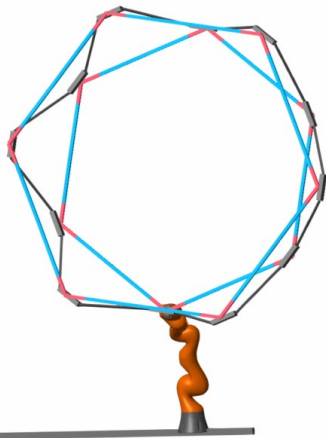




Introduction

Comparison of Robotic Manipulators and Avian Stabilization Systems

Category	Robotic Manipulator	Avian Stabilization System
Disturbances	Base vibrations; structural deformation	Body oscillations; gait-induced impacts
Objectives	End-effector stabilization; trajectory tracking	Head-pose stabilization; gaze stabilization
Actuation	Joint actuators; end-effector actuator	Multi-DOF neck; passive muscle elasticity
Sensing	IMU; infrared camera	Vestibular; visual systems



• On-Orbit Operations of a Flexible-Base Manipulator



• Stabilization



I

Introduction

II

Method

III

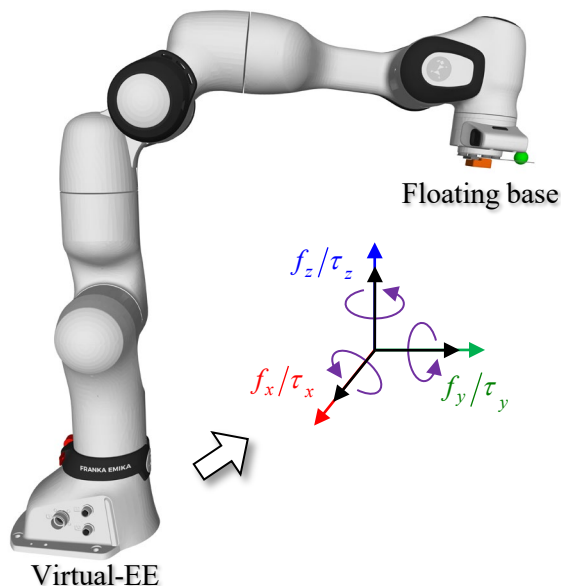
Results

IV

Conclusion and ongoing work

WBC Problem Formulation

Inverted robot model



Advantage

- Dynamic coupling aware;
Acceleration-level compensation.
- Robust to irregular disturbances;
- Naturally integrates with impedance control;
- Unifies multiple constraints within a single QP formulation.

★ High-fidelity dynamics model

Floating-Base dynamics

$$M(q)\ddot{q} + C(q, \dot{q}) + G(q) = \tau$$

$$\begin{bmatrix} M_{bb} & M_{bj} \\ M_{jb} & M_{jj} \end{bmatrix} \begin{bmatrix} \ddot{q}_b \\ \ddot{q}_j \end{bmatrix} + \begin{bmatrix} C_{bb} & C_{bj} \\ C_{jb} & C_{jj} \end{bmatrix} \begin{bmatrix} \dot{q}_b \\ \dot{q}_j \end{bmatrix} + \begin{bmatrix} G_b \\ G_j \end{bmatrix} = \begin{bmatrix} F_b \\ \tau_j \end{bmatrix} + J_g^T F_e$$

$$\downarrow \begin{bmatrix} C_{bb} & C_{bj} \\ C_{jb} & C_{jj} \end{bmatrix} \begin{bmatrix} \dot{q}_b \\ \dot{q}_j \end{bmatrix} + \begin{bmatrix} G_b \\ G_j \end{bmatrix} - J_g^T F_e = \begin{bmatrix} h_b \\ h_j \end{bmatrix}$$

$$\begin{cases} M_{bb}\ddot{q}_{b,ref} + M_{bj}\ddot{q}_j + h_b = F_b \\ M_{jb}\ddot{q}_{b,ref} + M_{jj}\ddot{q}_j + h_j = \tau_j \end{cases}$$

$$\downarrow \text{choose } x = [\ddot{q}_j^T, \tau_j^T]^T$$

$$\begin{bmatrix} M_{bj} & 0 \\ M_{jj} & -I \end{bmatrix} \begin{bmatrix} \ddot{q}_j \\ \tau_j \end{bmatrix} = \begin{bmatrix} -(h_b + M_{bb}\ddot{q}_{b,ref}) \\ -(h_j + M_{jb}\ddot{q}_{b,ref}) \end{bmatrix} \quad \text{Linear equality constraint}$$

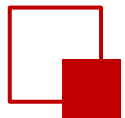
QP Formulation

- Joint virtual damping
- Dynamic equation
- CBF-based joint limit
- Motor torque

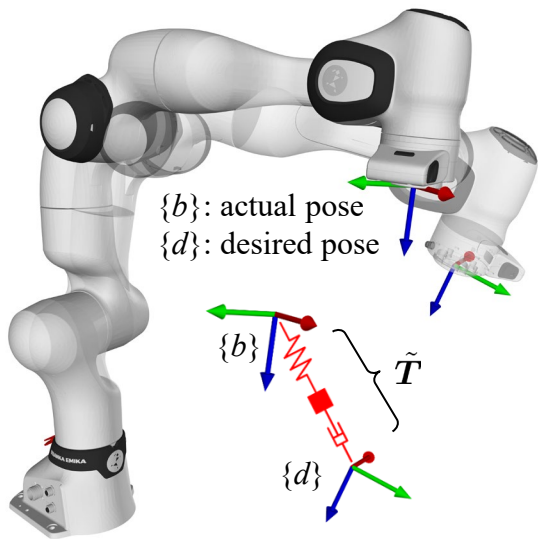
$$\begin{aligned} \min_{\ddot{q}_j, \tau_j} \quad & w_{\ddot{q}} \|\ddot{q}_j - \ddot{q}_{j,ref}\|^2 + w_{\tau} \|\tau_j - \tau_{j,ref}\|^2 \\ \text{s.t.} \quad & \begin{bmatrix} M_{bj} & 0 \\ M_{jj} & -I \end{bmatrix} \begin{bmatrix} \ddot{q}_j \\ \tau_j \end{bmatrix} = \begin{bmatrix} -(h_b + M_{bb}\ddot{q}_{b,ref}) \\ -(h_j + M_{jb}\ddot{q}_{b,ref}) \end{bmatrix} \\ & \begin{bmatrix} -I & 0 \\ I & 0 \\ 0 & -I \\ 0 & I \end{bmatrix} \begin{bmatrix} \ddot{q}_j \\ \tau_j \end{bmatrix} \leq \begin{bmatrix} k_1\dot{q}_j + k_0(q_j - q_{j,min}) \\ -k_1\dot{q}_j + k_0(q_{j,max} - q_j) \\ -\tau_{j,min} \\ \tau_{j,max} \end{bmatrix} \end{aligned}$$

Reference Input:

- Joint virtual damping: $\ddot{q}_{j,ref} = \frac{\kappa}{\Delta t} \dot{q}_j$
- Feedforward+SE(3) impedance: $\ddot{q}_{b,ref}$
- Minimize joint actuation effort: $\tau_{j,ref} = \mathbf{0}_7$



SE(3) Impedance-Based Motion Generation



• Impedance illustration

$$\begin{aligned} T &\in \text{SE}(3) \\ \left[T^{-1} \dot{T} \right] &\downarrow \\ V_b &\in \mathbb{R}^6 \\ \frac{d}{dt}(\cdot) &\downarrow \\ \dot{V}_b &\in \mathbb{R}^6 \end{aligned}$$

Body frame description with SE(3)

$$\begin{aligned} &\xleftrightarrow[\exp(\lambda)]{\log(T)} \\ &\xleftrightarrow[\text{dexp}_{-\lambda} \dot{\lambda}]{\text{dexp}_{-\lambda}^{-1} V_b} \\ &\xleftrightarrow[\text{dexp}_{-\lambda} \ddot{\lambda} + \left(\frac{d}{dt} \text{dexp}_{-\lambda} \right) \dot{\lambda}]{\text{dexp}_{-\lambda}^{-1} \left(\dot{V}_b - \left(\frac{d}{dt} \text{dexp}_{-\lambda} \right) \dot{\lambda} \right)} \end{aligned}$$

Exponential coordinate with se(3)

$$\begin{aligned} \lambda &\in \mathbb{R}^6 \\ \downarrow \frac{d}{dt}(\cdot) \\ \dot{\lambda} &\in \mathbb{R}^6 \\ \downarrow \frac{d}{dt}(\cdot) \\ \ddot{\lambda} &\in \mathbb{R}^6 \end{aligned}$$

Key Features

- Commutative Mapping between SE(3) and se(3)
- Desired Inertia Reshaping
- Geometrically Consistent Elastic Wrench
- Strict ODE in Lie Algebra

1) Impedance design on Lie group

$$A \dot{V} + D \tilde{V} + \text{dexp}_{\lambda}^{-T} K \lambda = \tilde{F}$$

- Specify desired inertia, damping, and stiffness

2) Impedance conversion to Lie algebra

$$A_{\lambda} \ddot{\lambda} + D_{\lambda} \dot{\lambda} + K_{\lambda} \lambda = \gamma$$

- Impedance ODE formulation in se(3)

3) Reference acceleration generation

$$\ddot{\lambda}_{\text{ref}} = -A_{\lambda}^{-1} D_{\lambda} \dot{\lambda} - A_{\lambda}^{-1} K_{\lambda} \lambda + A_{\lambda}^{-1} \gamma$$

- Generate $\ddot{\lambda}_{\text{ref}}$ from the se(3) impedance model

4) Lie group reference generation for control

$$\dot{V}_{\text{ref}} = \text{Ad}_{\tilde{T}} \dot{V}_d - \text{dexp}_{\lambda} \ddot{\lambda}_{\text{ref}} + \text{ad}_{\tilde{V}} V - \left(\frac{d}{dt} \text{dexp}_{\lambda} \right) \dot{\lambda}$$

- Generate $\dot{V}_{\text{ref}}$ for inverse dynamics control



I

Introduction

II

Method

III

Results

IV

Conclusion and ongoing work



I Introduction

II Method

III Results

IV Conclusion and ongoing work



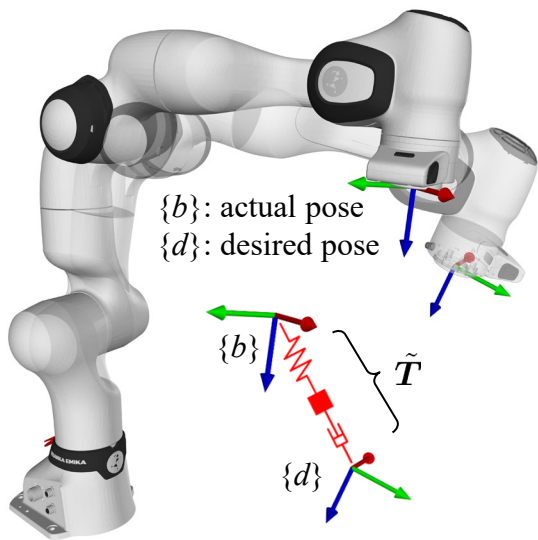
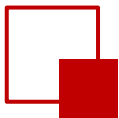
Conclusion and ongoing work

Conclusion:

- (1) A novel inverse-dynamics-based QP control framework with an inverted formulation is proposed, unifying stabilization and tracking tasks to achieve high-precision real-time control of the virtual end-effector.**
- (2) An SE(3) impedance method is employed to compensate for trajectory-tracking errors, improving tracking accuracy while endowing the framework with compliant interaction capabilities during manipulation tasks.**

Ongoing Work

- (1) The proposed framework has been preliminarily validated through experiments. Further validation and optimization are currently underway.**
- (2) An additional contact-rich manipulation experiment will need to be designed to validate the framework's compliant interaction capabilities during manipulation tasks.**
- (3) Dual-arm Cooperative Manipulation and Reactive Obstacle Avoidance (Future Work)**



• Impedance illustration

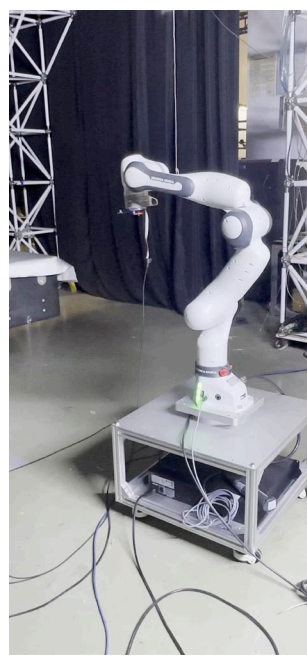
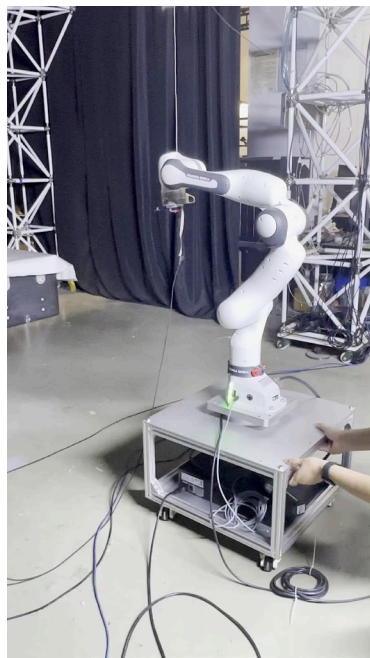
$$\begin{aligned}
 &T \in \text{SE}(3) \\
 &\left[T^{-1} \dot{T} \right] \downarrow \\
 &V_b \in \mathbb{R}^6 \\
 &\frac{d}{dt}(\cdot) \downarrow \\
 &\dot{V}_b \in \mathbb{R}^6
 \end{aligned}$$

Body frame description with SE(3)

$$\begin{aligned}
 &\xleftrightarrow[\exp(\lambda)]{\log(T)} \\
 &\xleftrightarrow[\text{dexp}_{-\lambda} \dot{\lambda}]{\text{dexp}_{-\lambda}^{-1} V_b} \\
 &\xleftrightarrow[\text{dexp}_{-\lambda} \ddot{\lambda} + \left(\frac{d}{dt} \text{dexp}_{-\lambda} \right) \dot{\lambda}]{\text{dexp}_{-\lambda}^{-1} \left(\dot{V}_b - \left(\frac{d}{dt} \text{dexp}_{-\lambda} \right) \dot{\lambda} \right)}
 \end{aligned}$$

$$\begin{aligned}
 &\lambda \in \mathbb{R}^6 \\
 &\downarrow \frac{d}{dt}(\cdot) \\
 &\dot{\lambda} \in \mathbb{R}^6 \\
 &\downarrow \frac{d}{dt}(\cdot) \\
 &\ddot{\lambda} \in \mathbb{R}^6
 \end{aligned}$$

Exponential coordinate with se(3)





北京理工大学
BEIJING INSTITUTE OF TECHNOLOGY

感谢聆听
请各位老师批评指正！

2026年07月15日